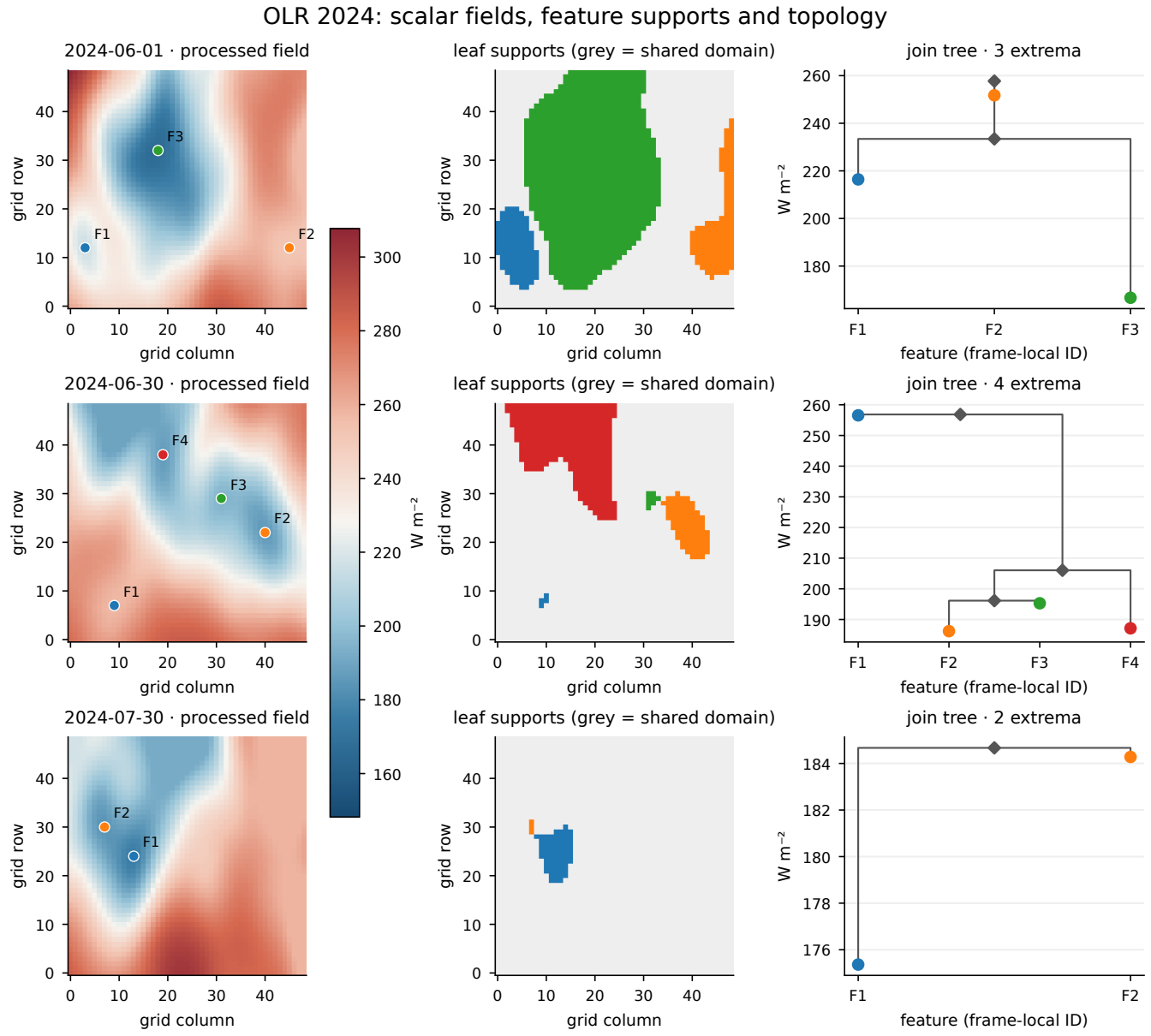
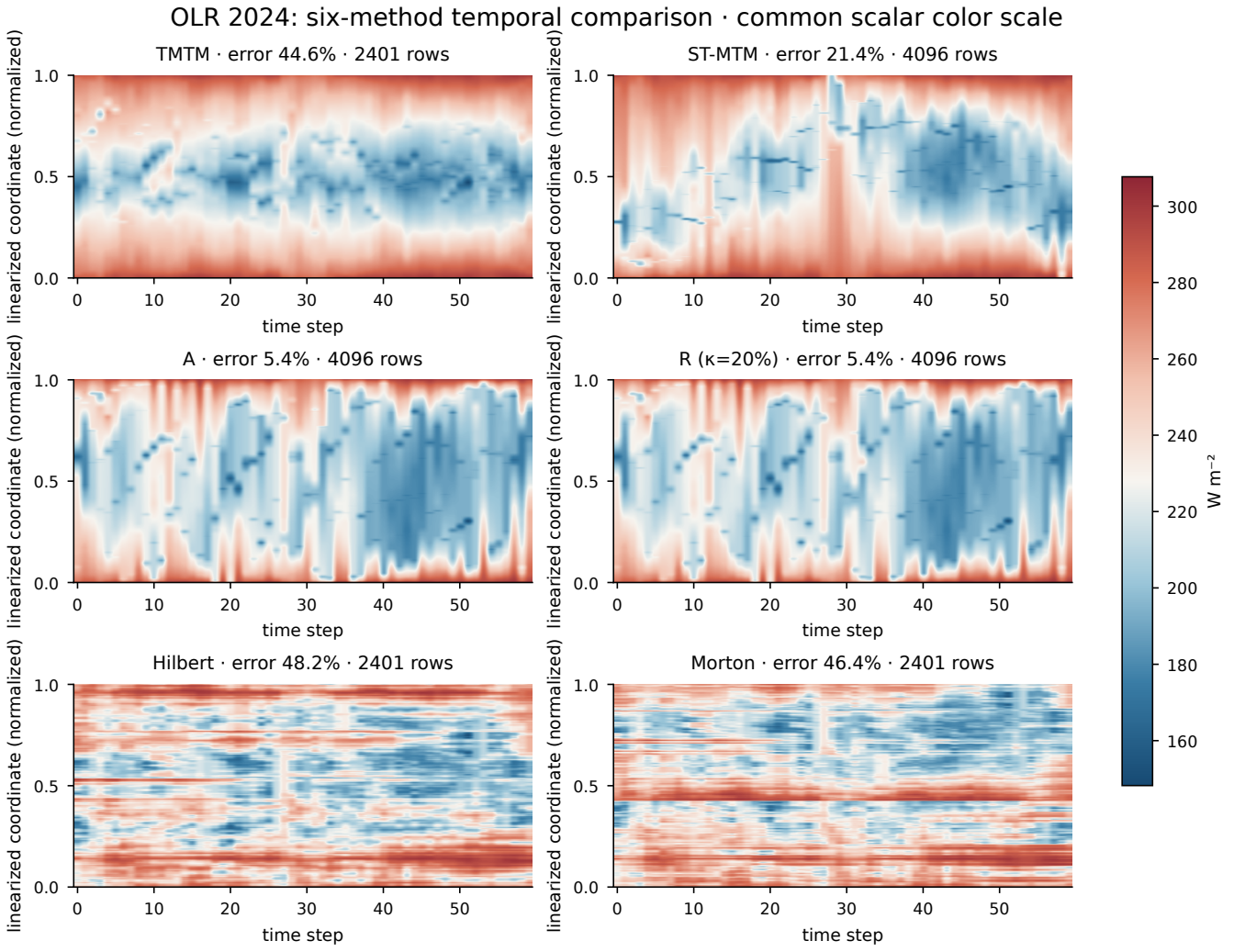
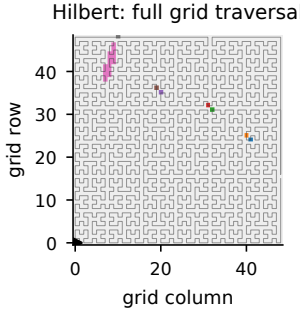
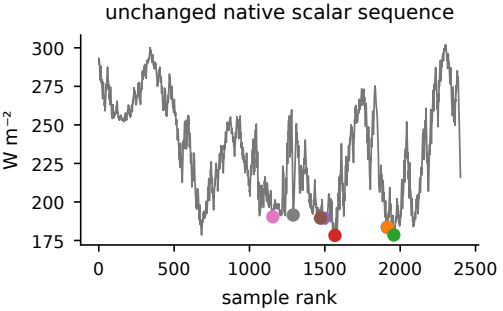
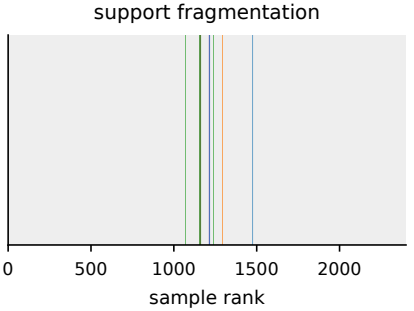

OLR 2024: scalar fields, feature supports and topology
2024-06-01 · processed field
leaf supports (grey = shared domain)
join tree · 3 extrema
2024-06-30 · processed field
leaf supports (grey = shared domain)
join tree · 4 extrema
2024-07-30 · processed field
leaf supports (grey = shared domain)
join tree · 2 extrema
grid row
grid column
W m⁻²
feature (frame-local ID)
F1
F2
F3
F4


OLR 2024: six-method temporal comparison · common scalar color scale
TMTM · error 44.6% · 2401 rows
ST-MTM · error 21.4% · 4096 rows
A · error 5.4% · 4096 rows
R (κ=20%) · error 5.4% · 4096 rows
Hilbert · error 48.2% · 2401 rows
Morton · error 46.4% · 2401 rows
linearized coordinate (normalized)
time step
0.0
0.5
1.0
0
10
20
30
40
50
W m⁻²
160
180
200
220
240
260
280
300

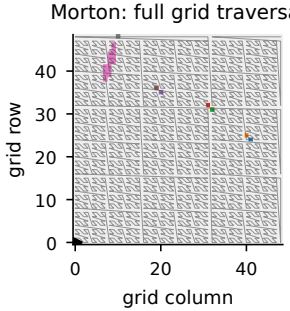
OLR 2024: native ordering and fragmentation · 2024-07-12

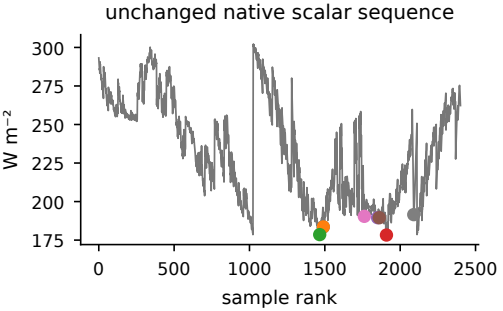


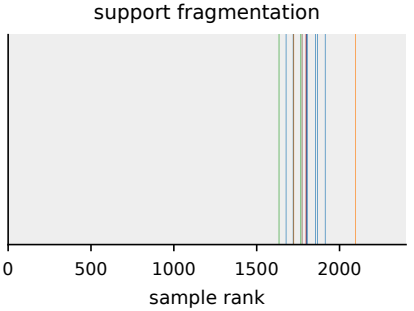




Runs: F1=1, F2=1, F3=1, F4=1
F5=1, F6=1, F7=6, F8=1

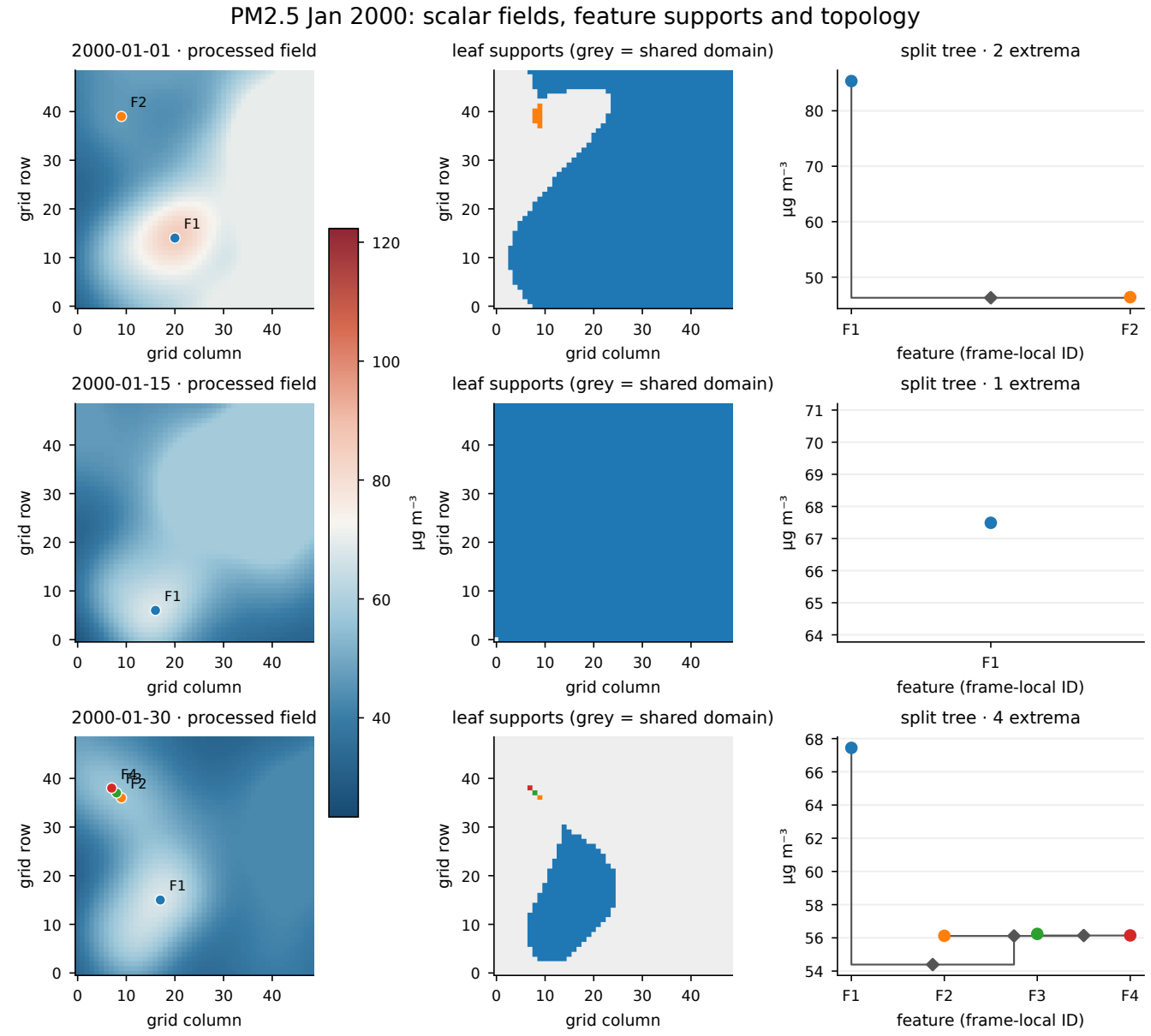


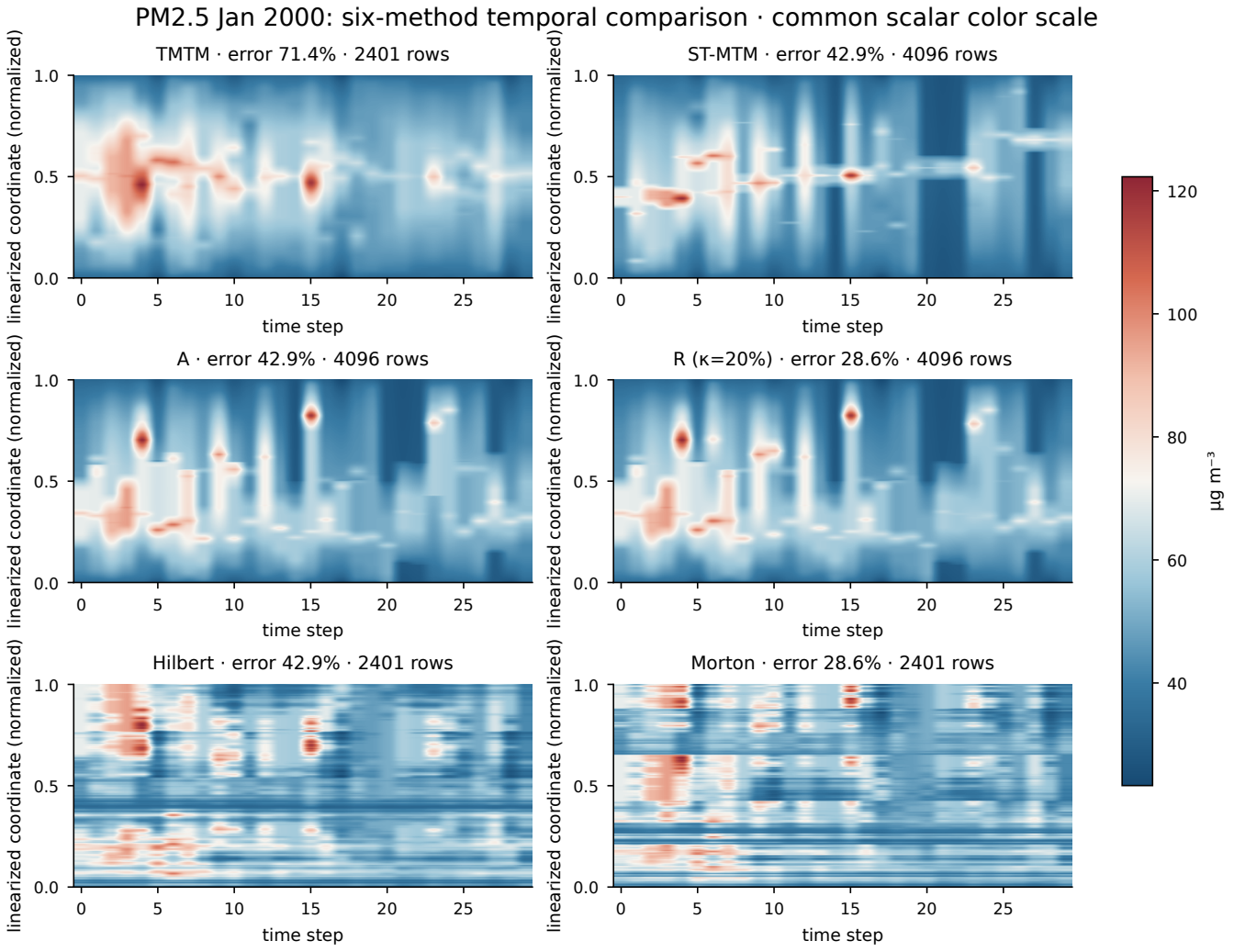


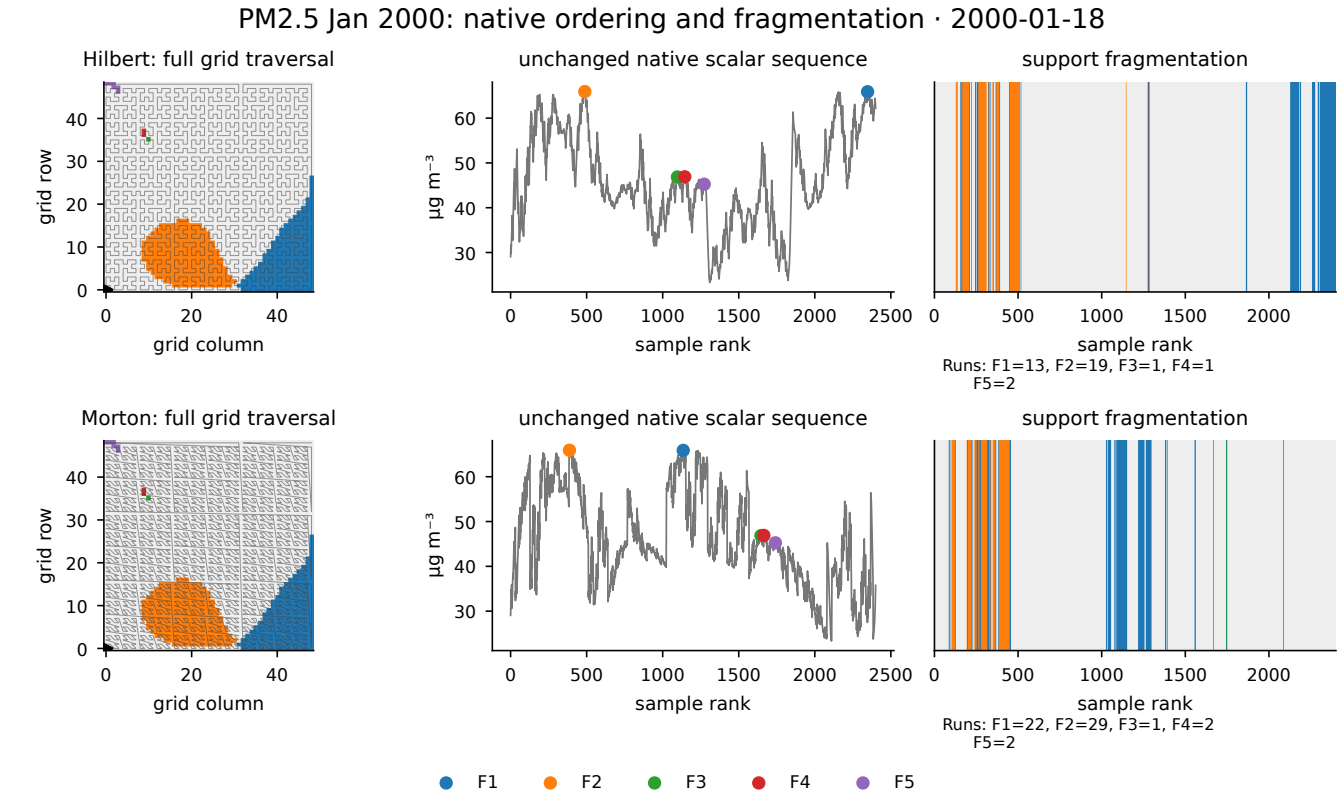


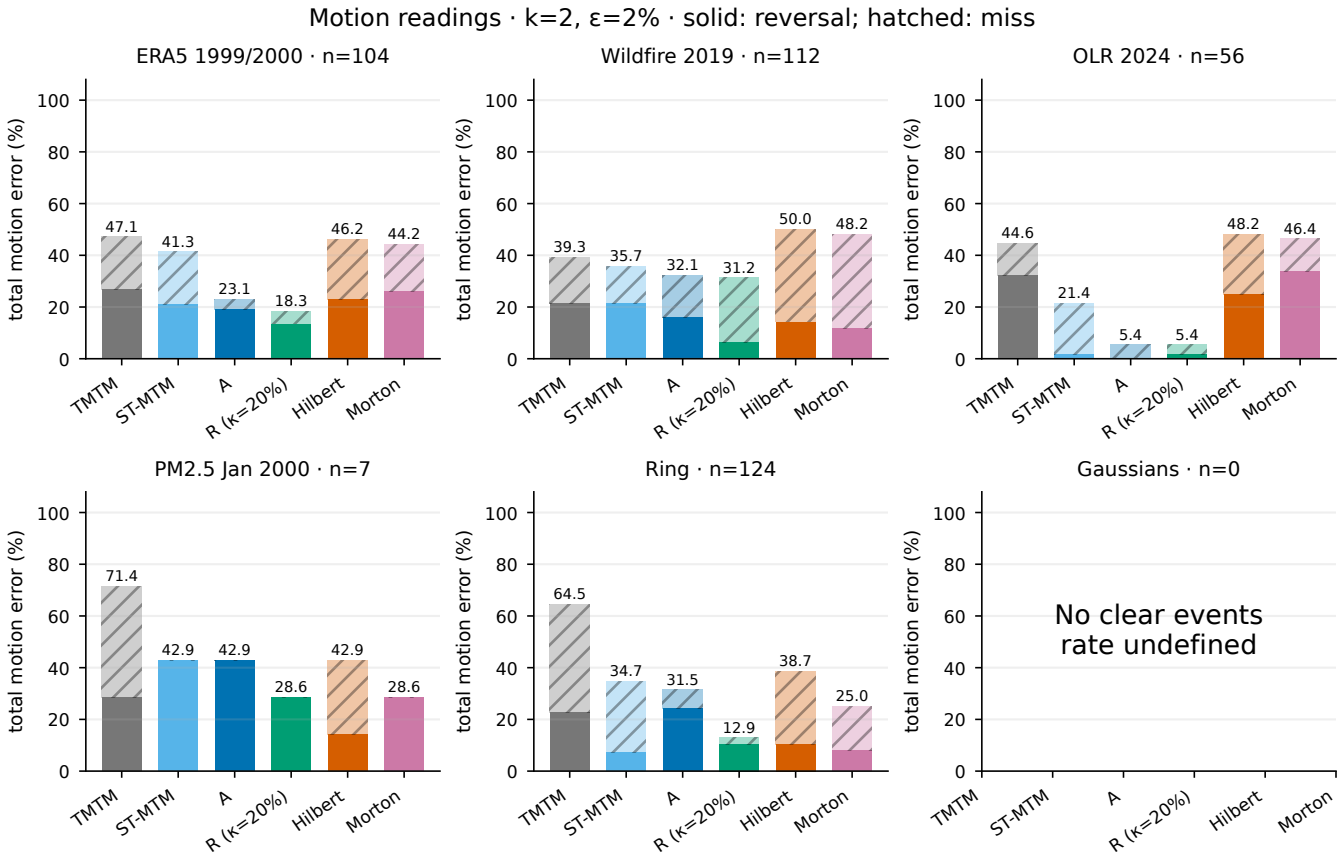
Runs: F1=1, F2=1, F3=1, F4=1
F5=1, F6=1, F7=11, F8=1

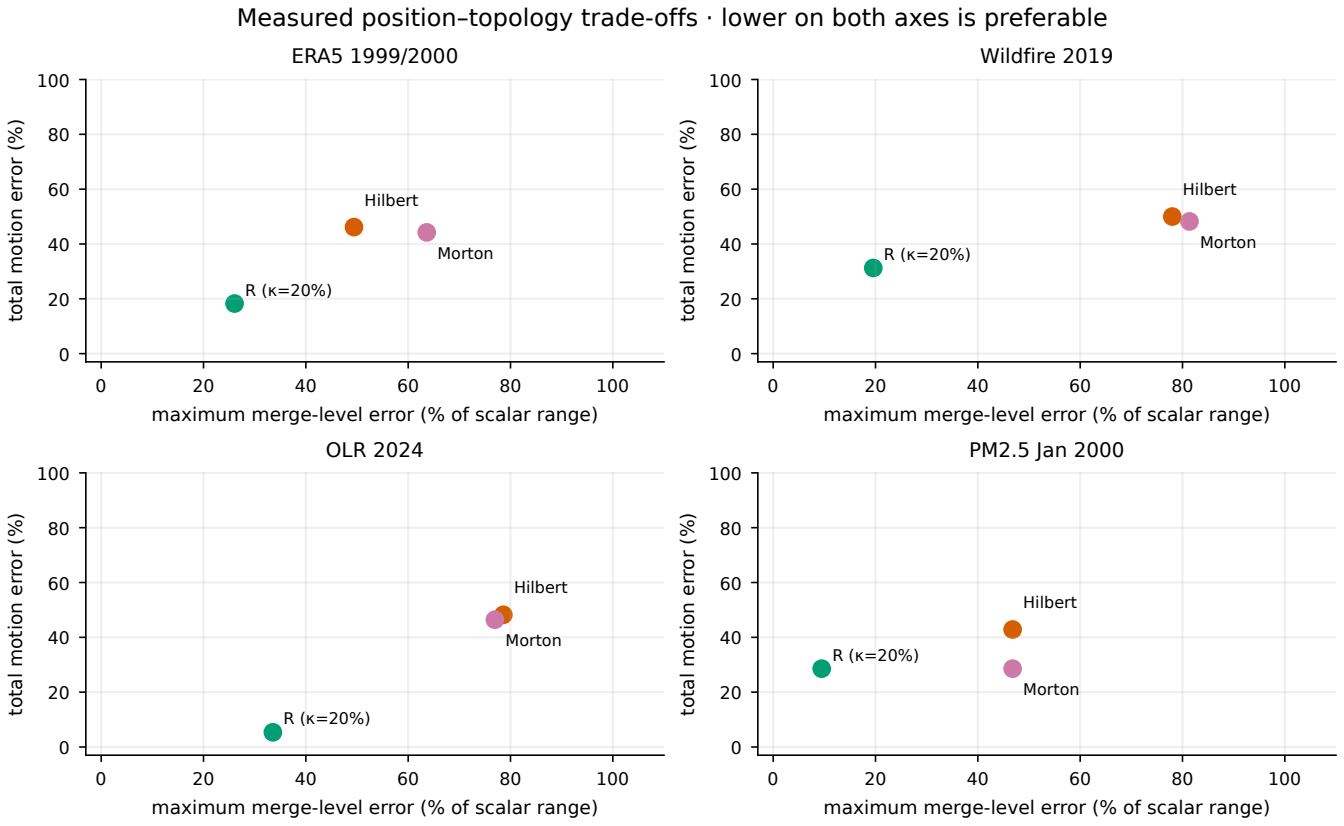
F1 F2 F3 F4 F5 F6 F7 F8

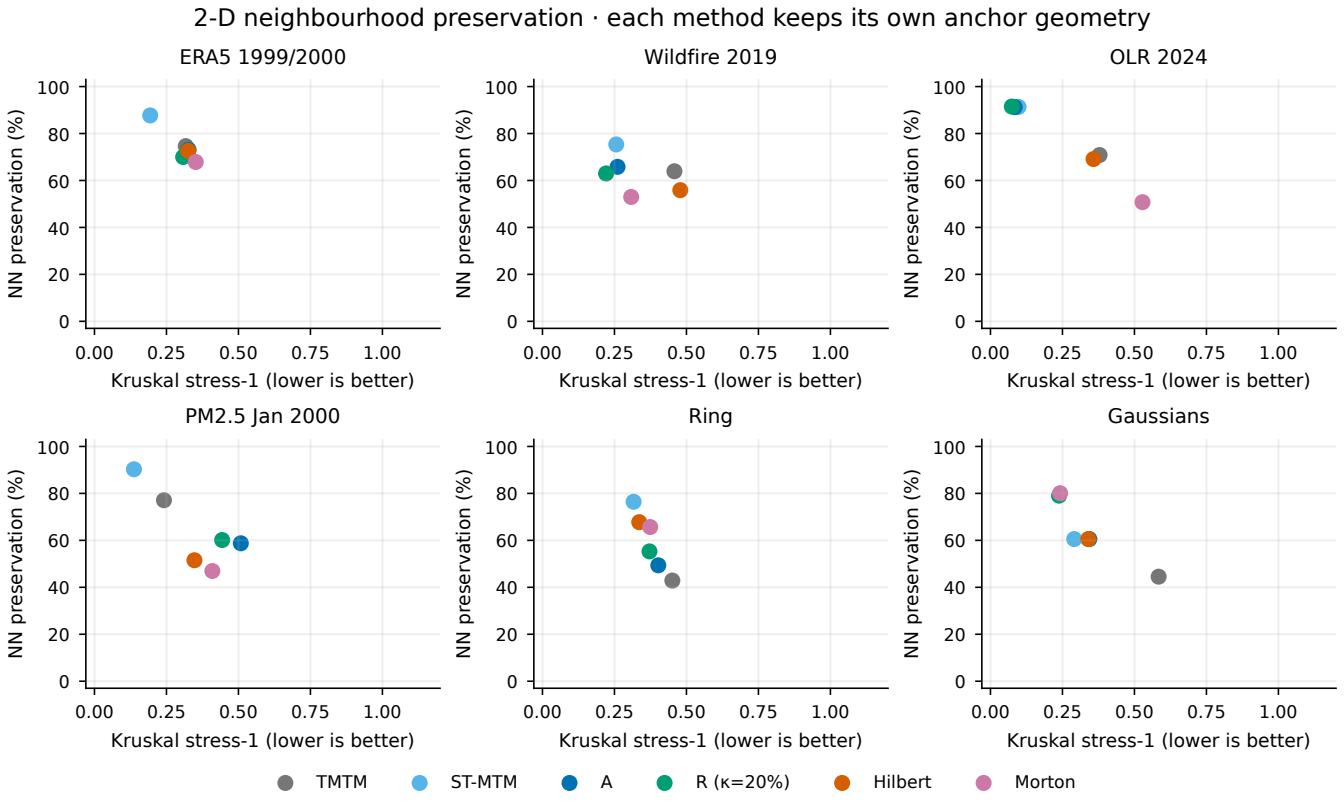

PM2.5 Jan 2000: scalar fields, feature supports and topology
2000-01-01 · processed field
leaf supports (grey = shared domain)
split tree · 2 extrema
2000-01-15 · processed field
leaf supports (grey = shared domain)
split tree · 1 extrema
2000-01-30 · processed field
leaf supports (grey = shared domain)
split tree · 4 extrema
grid row
grid column
µg m⁻³
feature (frame-local ID)
F1
F2
F3
F4

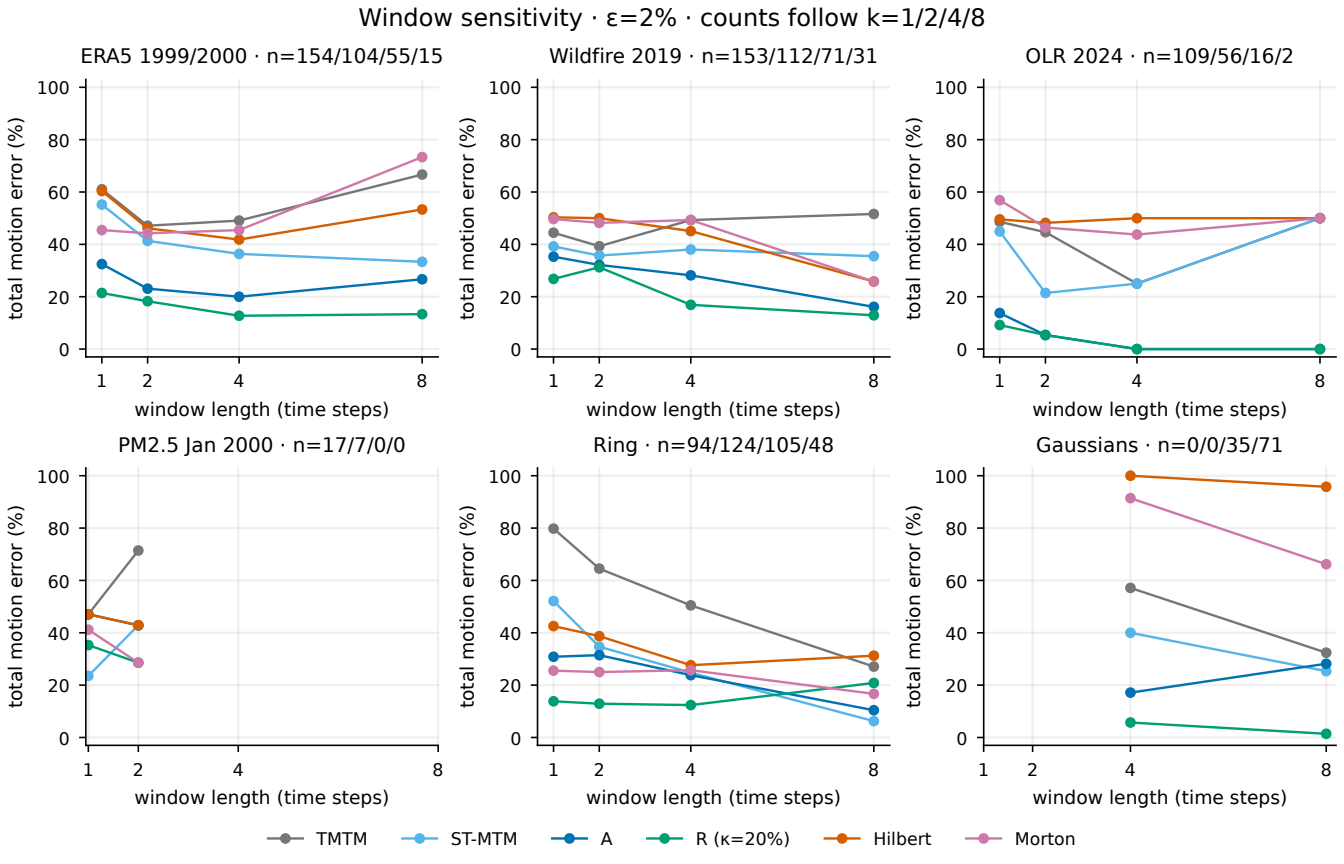

PM2.5 Jan 2000: six-method temporal comparison · common scalar color scale
TMTM · error 71.4% · 2401 rows
ST-MTM · error 42.9% · 4096 rows
A · error 42.9% · 4096 rows
R (κ=20%) · error 28.6% · 4096 rows
Hilbert · error 42.9% · 2401 rows
Morton · error 28.6% · 2401 rows
linearized coordinate (normalized)
time step
0.0
0.5
1.0
0
5
10
15
20
25
μg m⁻³
40
60
80
100
120


PM2.5 Jan 2000: native ordering and fragmentation · 2000-01-18
Hilbert: full grid traversal
grid row
grid column
unchanged native scalar sequence
µg m⁻³
sample rank
support fragmentation
sample rank
Runs: F1=13, F2=19, F3=1, F4=1
F5=2
Morton: full grid traversal
grid row
grid column
unchanged native scalar sequence
µg m⁻³
sample rank
support fragmentation
sample rank
Runs: F1=22, F2=29, F3=1, F4=2
F5=2
F1
F2
F3
F4
F5


Motion readings · k=2, ε=2% · solid: reversal; hatched: miss
ERA5 1999/2000 · n=104
Wildfire 2019 · n=112
OLR 2024 · n=56
PM2.5 Jan 2000 · n=7
Ring · n=124
Gaussians · n=0
total motion error (%)
0
20
40
60
80
100
47.1
41.3
23.1
18.3
46.2
44.2
39.3
35.7
32.1
31.2
50.0
48.2
44.6
21.4
5.4
5.4
48.2
46.4
71.4
42.9
42.9
28.6
42.9
28.6
64.5
34.7
31.5
12.9
38.7
25.0
No clear events
rate undefined
TMTM
ST-MTM
A
R (κ=20%)
Hilbert
Morton


Measured position–topology trade-offs · lower on both axes is preferable
ERA5 1999/2000
Wildfire 2019
OLR 2024
PM2.5 Jan 2000
total motion error (%)
maximum merge-level error (% of scalar range)
Hilbert
Morton
R (κ=20%)
0
20
40
60
80
100


2-D neighbourhood preservation · each method keeps its own anchor geometry
ERA5 1999/2000
Wildfire 2019
OLR 2024
PM2.5 Jan 2000
Ring
Gaussians
NN preservation (%)
Kruskal stress-1 (lower is better)
0
20
40
60
80
100
0.00
0.25
0.50
0.75
1.00
TMTM
ST-MTM
A
R (κ=20%)
Hilbert
Morton


Window sensitivity · ε=2% · counts follow k=1/2/4/8
ERA5 1999/2000 · n=154/104/55/15
Wildfire 2019 · n=153/112/71/31
OLR 2024 · n=109/56/16/2
PM2.5 Jan 2000 · n=17/7/0/0
Ring · n=94/124/105/48
Gaussians · n=0/0/35/71
total motion error (%)
window length (time steps)
0
20
40
60
80
100
1
2
4
8
TMTM
ST-MTM
A
R (κ=20%)
Hilbert
Morton